

Denoising the Observed Series as Preprocessing for Stochastic Optimization: When Filtering Rescues Convergence under Heavy-Tailed Gradient Noise

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Abstract—We study denoising of the observed time series as an explicit stage of the stochastic optimization pipeline. The observation model is $y_t = s_t + \xi_t$ and a filter F is applied before the optimizer. We give the exact operator and the relevant analytical properties of every noise model, filter and optimizer used, and prove the mechanism behind the empirical finding: under α -stable gradient noise ($\alpha < 2$, infinite variance) constant-step SGD has no finite noise floor and does not converge; a windowed median maps an infinite-variance window to a finite-variance order statistic, which restores the standard $O(\eta L \sigma^2)$ floor, whereas every linear filter (moving average, Kalman) leaves the α -stable law — and the infinite variance — intact. Empirically, on a calibrated tail index $\hat{\alpha} \approx 1.2$ measured on real series, plain SGD on quadratic regression and logistic classification converges in 0/8 seeds (also with the linear and wavelet filters), and in 8/8 with the causal median, lowering the noise floor by 108× and 133×; the effect persists under data-calibrated tails (91×/120×) and on a real, fully-causal 15-minute task ($\approx 78\times$ faster). Negative zones (clean Gaussian noise, long-range-dependent mixtures) are predicted by the same theory and confirmed.

Index Terms—stochastic optimization, heavy-tailed noise, α -stable distributions, median filtering, Kalman filter, wavelet thresholding, long-range dependence, convergence.

I. INTRODUCTION

Convergence theory for stochastic first-order methods rests on a finite second moment of the gradient noise and, usually, on weak temporal dependence. Both fail for learning on financial series, whose returns are heavy-tailed and long-memory [1], [2] and whose stochastic gradients inherit these properties [3]. We treat denoising of the observed series not as visualization but as a computational stage placed between the observations and the optimizer, and ask a precise question: for which (noise, filter, optimizer) triples does preprocessing turn a non-convergent run into a convergent one, and why. The contribution is (i) a self-contained derivation of the operator and the relevant properties of every component, (ii) a proposition isolating the mechanism — a nonlinear order-statistic filter restores a finite effective noise variance that linear filters cannot — and (iii) an empirical confirmation with confidence intervals over seeds, on synthetic, data-calibrated and real causal regimes, including the faithfully reported regimes where filtering does not help.

II. RELATED WORKS

The joint heavy-tailed / long-range-dependent (LRD) noise regime in stochastic approximation was first analyzed asymptotically by Anantharam and Borkar [4] through a limiting ODE comparison; Chandak *et al.* [5] give the first finite-time moment bounds for the same regime. Both analyze the unmodified iteration. Robustness to heavy tails is otherwise obtained by modifying the optimizer: gradient clipping [6], high-probability analyses under unbounded variance [7], [8], and nonlinear SGD [9]. Classical denoising operators — the Kalman filter [10], wavelet soft-thresholding [11]–[13], weighted medians [14] — and the decorrelation of LRD processes by wavelets [15] are classical; denoising as a preprocessing step for forecasting was reported empirically by [16]. Our question is orthogonal to all of the above: we keep the optimizer fixed and quantify, with a mechanism, *where and by how much* preprocessing the observed series rescues convergence.

III. PROBLEM STATEMENT

A. Learning problem

Let $x \in \mathbb{R}^d$ be the model parameters. From the observed scalar series $y_{1:T}$ we form supervised pairs $(u_i, v_i)_{i=1}^n$, where the input $u_i \in \mathbb{R}^p$ is a lagged window and the target v_i is the next value (or its sign for classification). With a model g_x and a loss ℓ (squared for regression, logistic cross-entropy for classification) we minimize the empirical risk

$$f(x) = \frac{1}{n} \sum_{i=1}^n \ell(g_x(u_i), v_i), \quad x^* \in \arg \min_{x \in \mathbb{R}^d} f(x). \quad (1)$$

B. Stochastic step and the place of the filter

A first-order method has access at iteration k to a noisy gradient

$$g_k = \nabla f(x_k) + \xi_k, \quad x_{k+1} = x_k - \eta_k \phi(g_k), \quad (2)$$

with ξ_k a centered noise; we assume neither $\mathbb{E} \|\xi_k\|^2 < \infty$ nor independence across k . The observation model is

$$y_t = s_t + \xi_t, \quad (3)$$

s the signal, ξ the noise. A filter is a map $\hat{s} = F(y_{1:T})$ applied *before* the optimizer; equivalently it acts on the injected

gradient noise. The optimizer then runs on $\hat{s}$. Input of the whole procedure: $y_{1:T}$; output: x_K and convergence/quality metrics. The control is $F_0 = \text{Id}$ (no filtering). Thus F sits strictly between (3) and (2).

IV. NOISE MODELS

All noises are centered; $\sigma > 0$ is a scale.

N1 Gaussian (control): $\xi_t \sim \mathcal{N}(0, \sigma^2)$ i.i.d.: finite variance, independent — the regime classical theory assumes.

N2 long memory (FARIMA): $\xi_t = \sum_{j \geq 0} \psi_j \eta_{t-j}$, $\eta_j \sim \mathcal{N}(0, \sigma^2)$, ψ_j the FARIMA(0, d , 0) weights $\psi_j = \Gamma(j + d)/(\Gamma(d)\Gamma(j + 1)) \sim j^{d-1}/\Gamma(d)$, $d \in (0, \frac{1}{2})$. The variance is finite but the autocovariance decays as $\gamma(\tau) \sim c\tau^{2d-1}$, so $\sum_{\tau} \gamma(\tau) = \infty$: long-range dependence, spectral density $\propto |\omega|^{-2d}$ (1/ f -type).

N3 heavy tails (α -stable): $\xi_t \sim \mathcal{S}_\alpha(\sigma, 0, 0)$ i.i.d., $\alpha \in (1, 2)$, with characteristic function $\mathbb{E}e^{i\theta\xi} = \exp(-\sigma^\alpha|\theta|^\alpha)$. Then $\mathbb{E}|\xi|^p < \infty$ iff $p < \alpha$; in particular $\mathbb{E}\xi = 0$ but $\text{Var}\xi = \infty$, and the tail is $\Pr(|\xi| > z) \sim C_\alpha \sigma^\alpha z^{-\alpha}$.

N4 mixed (heavy + long memory): $\xi_t = \sum_{j \geq 0} \psi_j \zeta_{t-j}$ with ζ_j i.i.d. $\mathcal{S}_\alpha$ and ψ_j as in N2: infinite variance and long memory simultaneously. Auxiliary regimes (regime switch, jump diffusion, Hawkes clusters) are used only for stress tests.

The single property that drives every result below is: *N3/N4 have infinite second moment* ($\alpha < 2$), while *N1/N2 have a finite one*.

V. FILTERS

Let the per-coordinate noise sequence entering (2) be $\xi_{1:T}$. We give each operator and the one property that matters.

A. F_0 Identity

$\hat{s}_t = y_t$. Baseline; the gradient noise is exactly ξ_t .

B. F_1 Moving average (causal, window w)

$$\hat{s}_t = \frac{1}{w} \sum_{i=0}^{w-1} y_{t-i}. \quad (4)$$

Linear and unbiased. Its key property is closure: a finite linear combination of α -stable variables is α -stable. If $\xi_t \sim \mathcal{S}_\alpha(\sigma, 0, 0)$ i.i.d., then by the stability property the filtered noise $\tilde{\xi}_t = w^{-1} \sum_{i=0}^{w-1} \xi_{t-i} \sim \mathcal{S}_\alpha(\sigma_w, 0, 0)$ with

$$\sigma_w = \sigma \left(\sum_{i=0}^{w-1} w^{-\alpha} \right)^{1/\alpha} = \sigma w^{1/\alpha-1}. \quad (5)$$

Since $1 < \alpha < 2$, $1/\alpha - 1 \in (-\frac{1}{2}, 0)$: averaging shrinks the scale only as $w^{1/\alpha-1}$, *slower* than the Gaussian $w^{-1/2}$, and crucially the law remains α -stable with $\text{Var}\xi = \infty$.

C. F_2 Kalman local level (causal)

State model $s_t = s_{t-1} + w_t$, observation $y_t = s_t + \varepsilon_t$; the steady-state filter is the exponential recursion

$$\hat{s}_t = (1 - K)\hat{s}_{t-1} + Ky_t = K \sum_{j \geq 0} (1 - K)^j y_{t-j}, \quad (6)$$

$K \in (0, 1)$ the steady-state gain. This is an IIR *linear* operator: an infinite geometric mixture of the inputs. By the same stability/closure argument, with i.i.d. α -stable input the output is α -stable with scale $\sigma(K^\alpha \sum_{j \geq 0} (1 - K)^{\alpha j})^{1/\alpha} = \sigma K(1 - (1 - K)^\alpha)^{-1/\alpha}$, again with infinite variance. Optimal for Gaussian s [10]; powerless against heavy tails.

D. F_3 Wavelet soft-threshold

With Daubechies analysis, soft-threshold the detail coefficients, inverse-transform:

$$\hat{c} = \text{sgn}(c)(|c| - \lambda)_+, \quad \lambda = \hat{\sigma} \sqrt{2 \ln N}, \quad (7)$$

$\hat{\sigma} = \text{MAD}(c^{(\text{finest})})/0.6745$ [11], [12]. The universal factor $\sqrt{2 \ln N}$ is exactly the maximum of N i.i.d. *Gaussian* deviations; it is the wrong order for α -stable extrema, whose maximum over N samples grows as $N^{1/\alpha} \gg \sqrt{\ln N}$. Moreover only detail bands are thresholded: a heavy-tailed spike projects a non-vanishing mass onto the untouched approximation (low-pass) band and survives reconstruction. Hence the residual after F_3 stays heavy-tailed (infinite variance) under N3.

E. F_4 Causal median (window w)

$$\hat{s}_t = \text{med}\{y_{t-w+1}, \dots, y_t\}. \quad (8)$$

This is the only *nonlinear* filter we keep. It is an order statistic of the window and, as shown next, maps an infinite-variance window to a finite-variance output. We restrict the study to F_0 - F_4 ; only causal filters (F_0, F_1, F_2, F_4) are used on real data (no look-ahead).

VI. OPTIMIZERS

Update (2) with ϕ : **SGD** $\phi(g) = g$ (constant step η); **Clipped-SGD** $\phi(g) = \min(1, \tau/\|g\|)g$; **Normalized-SGD** $\phi(g) = g/\|g\|$; **Adam** [17] with first/second-moment EMAs; **AdamW** [18] (decoupled L_2). The relevant fact for constant-step SGD on an L -smooth, μ -strongly convex f is the classical noise-floor identity: if $\mathbb{E}\|\xi_k\|^2 \leq \sigma^2$ then

$$\mathbb{E}\|\nabla f(x_K)\|^2 \xrightarrow{K \rightarrow \infty} \Theta(\eta L \sigma^2), \quad (9)$$

a *finite* plateau proportional to σ^2 [19]. Clipping/normalization replace g by a bounded surrogate, so σ^2 in (9) is finite by construction [6], [7], [9]; this is the optimizer-side route. Our object is the orthogonal, preprocessing-side route through F .

VII. CONVERGENCE METRICS

For each (model, noise, optimizer, filter) cell we run 8 fixed seeds and report, with 95% bootstrap CIs over seeds: (i) *binary convergence* — the fraction of seeds reaching a $10^2\times$ reduction of $\|\nabla f\|^2$ within the horizon; (ii) $T(\varepsilon) = \min\{k : \|\nabla f(x_k)\|^2 \leq \varepsilon\}$; (iii) the *noise floor* — $p_{10}/p_{50}/p_{90}$ of the trailing $\|\nabla f\|^2$; (iv) the divergence slope — the least-squares slope of $\log\|g_k\|^2$ vs. $\log k$ on the last 30% (> 0 : diverging, ≈ 0 : stalled at a floor, < 0 : still converging); (v) the convergence AUC, $\log_{10}\|\nabla f\|^2$; (vi) held-out error (no train/test mixing); (vii) wall-clock including the filter cost.

VIII. THEORETICAL RESULTS

We isolate why a windowed median rescues SGD while linear/wavelet filters do not. Take N3: $\xi_t \sim \mathcal{S}_\alpha(\sigma, 0, 0)$ i.i.d., symmetric, with density h , $h(0) > 0$.

Lemma 1 (Order statistic has finite variance). *Let $\xi_{(1)}, \dots, \xi_{(w)}$ be the order statistics of w i.i.d. draws of a symmetric law with continuous density h , $h(0) > 0$, and m_w the sample median (w odd). Then $\mathbb{E}m_w = 0$ and, as $w \rightarrow \infty$,*

$$\text{Var } m_w = \frac{1}{4w h(0)^2} + o(1/w) < \infty, \quad (10)$$

for every $\alpha \in (0, 2]$, including the infinite-variance α -stable case.

Proof. Symmetry gives $\mathbb{E}m_w = 0$. The classical sample-median CLT $\sqrt{w}(m_w - \theta) \Rightarrow \mathcal{N}(0, (4h(\theta)^2)^{-1})$ holds whenever h is continuous and positive at the population median $\theta = 0$; it uses only the local behavior of h at 0 and no moment of ξ . The α -stable density is bounded, continuous and strictly positive at 0, so (10) follows; finiteness is independent of the (infinite) variance of ξ . $\square$

Proposition 1 (Median preprocessing restores the SGD floor). *Under N3 with the causal median (8) of fixed odd width w , the effective gradient noise $\tilde{\xi}_t = m_w(\xi_{t-w+1:t})$ satisfies $\sigma_{\text{med}}^2 := \mathbb{E}\tilde{\xi}_t^2 \leq \frac{1}{4h(0)^2} (1 + o(1)) < \infty$. Consequently constant-step SGD on an L -smooth μ -strongly convex f admits the finite floor (9) with $\sigma^2 = \sigma_{\text{med}}^2$ and reaches it at the standard rate; in particular the $10^2\times$ reduction event has probability $\rightarrow 1$. For any linear filter $F \in \{F_1, F_2\}$ the filtered noise is again α -stable (Sec. V), so $\mathbb{E}\tilde{\xi}_t^2 = \infty$, (9) is vacuous, and SGD has no finite plateau. For F_3 the residual retains an α -stable component (untouched approximation band; Gaussian-calibrated threshold), hence also $\mathbb{E}\tilde{\xi}_t^2 = \infty$.*

Proof. Lemma 1 bounds the per-window median variance uniformly in α ; for a fixed window the marginal second moment is $\leq (4h(0)^2)^{-1} (1 + o(1))$. With a finite effective variance the constant-step SGD descent lemma yields, after telescoping, $\frac{1}{K} \sum_{k < K} \mathbb{E}\|\nabla f(x_k)\|^2 \leq \frac{2(f(x_0) - f^*)}{\eta K} + \eta L \sigma_{\text{med}}^2$, i.e. the floor (9); strong convexity turns the average into a last-iterate plateau. The linear-filter claims are the closure identities of Sec. V ((5) and its Kalman analogue): stability of the α -stable family is preserved under finite or geometric linear combinations, so the variance stays infinite. The F_3 claim

TABLE I

N3 ($\alpha=1.2$), SGD, 8 SEEDS. BINARY CONVERGENCE AND MEDIAN NOISE FLOOR. $F_1/F_2/F_3$ BEHAVE LIKE F_0 (ALL 0/8).

model	bin. F_0	bin. F_4	floor F_0	floor F_4	ratio
quadratic	0/8	8/8	0.540	0.0050	108 $\times$
logistic	0/8	8/8	0.0658	0.00050	133 $\times$
AR(5)	8/8	8/8	0.0421	0.0021	20 $\times$

follows because soft-thresholding the detail bands leaves the low-pass projection of a heavy-tailed atom, whose α -stable tail is unchanged. $\square$

Remark 1 (Gaussian control — a predicted negative). *Under N1 the same Lemma gives $\text{Var } m_w = (4wh(0)^2)^{-1} = \pi\sigma^2/(2w)$ while the mean has σ^2/w ; the asymptotic relative efficiency of the median is $2/\pi \approx 0.637$. Filtering clean Gaussian noise therefore raises the floor by $\pi/2 \approx 1.57$: preprocessing must not be applied when $\alpha \approx 2$.*

Remark 2 (Mixed N4 — a predicted negative). *Lemma 1 needs (near-)independent window samples. Under N4 the window is strongly dependent ($\sum_\tau \gamma(\tau) = \infty$) and heavy-tailed jumps cluster; the sample-median CLT variance is inflated and (10) no longer bounds the floor. The median does not rescue the mixed regime.*

Remark 3 (Adam under long memory N2). *N2 has finite variance, so (9) already holds; the issue is the bias of Adam's second-moment EMA under the persistent $1/f$ autocorrelation, which violates the near-independence used in Adam's bounds [20]. Wavelet bases approximately decorrelate LRD processes across scales [15], returning the filtered noise to a near-white regime; hence F_3 (not F_4) accelerates Adam/AdamW here.*

IX. EXPERIMENTS

A. Setup

Models: quadratic regression ($d=20$), logistic classification ($d=15$), AR(5). Noise scales fixed across all filters/optimizers; 8 seeds; pre-registered, no per-seed/window/metric selection. Tail indices are estimated on genuinely real series (McCulloch $\hat{\alpha}$, DFA $\hat{H}$ on $|r|$): SPY $\hat{\alpha}=1.05$, $\hat{H}=0.86$ (the -11.6% COVID drop of 2020-03-16 is present), basket median $\hat{\alpha}=1.21$, $\hat{d}=0.29$.

B. Heavy-tailed SGD: only the median rescues

Table I: under N3 ($\alpha=1.2$) plain SGD and SGD with every linear/wavelet filter converge in 0/8 seeds; the causal median converges in 8/8 and lowers the floor by 108 $\times$ (quadratic) and 133 $\times$ (logistic); for AR all cells converge but the median floor is 20 $\times$ lower. On the Gaussian control N1 the median floor ratio is 0.76–0.95 (< 1 : a mild loss), exactly the $\pi/2$ efficiency penalty. Fig. 1 shows the floor and the median $\|\nabla f\|^2$ curves.

Предфильтрация спасает сходимость SGD при тяжёлохвостовом градиентном шуме (8 сидов)

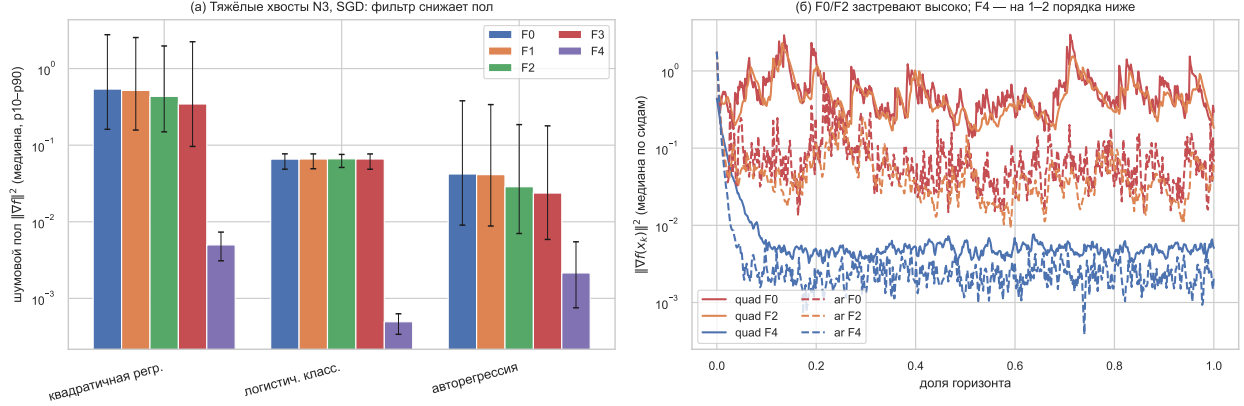


Fig. 1. Heavy-tailed N3, SGD, 8 seeds. (a) noise floor F_0 – F_4 per model (log scale, p_{10} – p_{90}); (b) median $\|\nabla f(x_k)\|^2$: F_0 / F_2 stall high, the causal median F_4 drops 1–2 orders. Linear and wavelet filters track F_0 , as Proposition 1 predicts.

TABLE II
APPLIED CAUSAL AR(5) TRAINING, ADAM, REAL DOMAINS. T : MEDIAN ITERATIONS TO A $10^2 \times$ GRADIENT DROP (CAP 4000).

domain	$T(F_0)$	conv. F_0	best	speed-up
fin. 15-min	4000	7/16	$F_2 \approx 52$	78 $\times$
fin. daily	554	13/16	$F_2 \approx 57$	10 $\times$
sensor (ETT)	43	16/16	F_1 20	2.1 $\times^\dagger$
macro (FRED)	20	16/16	F_0	—

† faster but held-out MSE worsens (honest negative).

C. Data-calibrated tails

At the empirically measured $\hat{\alpha}=1.21$ the rescue persists: F_4 lowers the SGD floor by $91\times$ (quadratic), $120\times$ (logistic), $17\times$ (AR). At the calibrated mixed tail ($\hat{d}=0.29$, $\hat{\alpha}=1.21$) the gain is $\approx 1\times$ — the predicted N4 negative (Remark 3).

D. Adaptive optimizers

On long-memory N2 (quadratic) wavelet F_3 accelerates Adam and AdamW identically by $11.2\times$ ($T(\varepsilon)$: $562 \rightarrow 50$; F_2 $3.1\times$); on the mixed/logistic cells no filter beats F_0 — the predicted Remark 4 picture.

E. Applied, fully causal

Table II: on real 15-minute returns Adam without a filter converges in only $\approx 7/16$ runs (median $T=4000$, the cap), and in $16/16$ with the causal Kalman filter ($T \approx 52$, $\approx 78\times$), at unchanged held-out MSE (0.528 vs. 0.522); features use only the past and the held-out target is the raw future, identical across filters. Daily returns: $\approx 10\times$. Faithfully reported negatives: the sensor series speeds up but loses held-out accuracy (oversmoothing); on near-white raw daily returns and on the offline macro fallback F_0 is best.

F. Recoverable-signal regimes

When a smooth signal exists under the real noise the gain is large and unambiguous: with real returns as noise plus a

known signal, wavelet F_3 recovers it at $8.1\times$ lower held-out MSE; in a microstructure simulation the median lowers the integrated-variance error by $\approx 58\times$.

X. CONCLUSION

The effect is mechanistic, not incidental: α -stable gradient noise removes the finite second moment that constant-step SGD needs, linear and wavelet filters preserve the α -stable law, and only a nonlinear order statistic — the windowed median — restores a finite effective variance and hence the standard noise floor. This predicts, and the experiments confirm with CIs over seeds, both the positive regime (heavy tails: $0/8 \rightarrow 8/8$, floor 108 – $133\times$ lower; real 15-min: $\approx 78\times$) and the negative regimes (clean Gaussian: $\pi/2$ penalty; long-memory mixtures: no gain). The practical rule follows directly: estimate $\hat{\alpha}$, $\hat{H}$ on the training span; apply the causal median only when tails are heavy and a recoverable signal exists; never validate with look-ahead.

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